Kellen Kanarios

University of Michigan, Ann Arbor, MI
kellenkk@umich.edu | +1 (917) 975-6415 | www.linkedin.com/in/KellenKanarios | Personal Website

EDUCATION

University of Michigan, Ann Arbor, MI — August 2024 | Present
Ph.D. Candidate in Electrical and Computer Engineering (in progress)

University of Michigan, Ann Arbor, MI — August 2024 | May 2026
M.S. in Electrical and Computer Engineering, Signal and Image Processing and Machine Learning — GPA: 3.9/4.00

University of Michigan, Ann Arbor, MI — August 2020 | August 2024
B.S. in Mathematics (with Honors) and Computer Science (with High Honors) — GPA: 3.85/4.00
Thesis Title: *Meta Learning for Continual Reinforcement Learning* (link)

SKILLS

- **Programming:** Python, C, C++, CUDA, LaTeX
- **ML/Systems:** JAX, PyTorch, XLA, Pallas
- **Software:** Git, Vim, Linux, Tmux

RESEARCH EXPERIENCE

PhD Researcher — Ann Arbor, MI
University of Michigan — August 2024 | Present

- Implemented a novel unsupervised reinforcement learning + tree search framework for high-dimensional motion planning.
- Wrote entire inference pipeline in JAX, enabling efficient root node parallelization via *just-in-time* compilation.
- Achieved 100% success rate on obstacle avoidance in high-dimensional Mujoco robotics environments.

Research Intern — Ann Arbor, MI
University of Michigan SURE Program — May 2024 | August 2024

- Proposed cost-aware bandit framework minimizing heterogeneous sampling cost under fixed-confidence guarantees.
- Derived cost complexity bound and designed both a provably asymptotically optimal (CTAS) and efficent (CO) algorithm.
- Showed that cost-aware exploration significantly outperforms standard methods in realistic settings accounting for cost.

High Performance Computing Intern — Los Angeles, CA
Lawrence Livermore National Laboratory — June 2023 | August 2023

- Designed new parallelizable block relaxation + coarsening strategy for solving higher-order PDE that arises in fusion.
- Demonstrated our method requires $\approx \mathbf{10^8}\times$ **fewer iterations** than algebraic multigrid to reach desired convergence rate.
- Selected 1 of 36 from 5,000+ applicants for UCLA's Research in Industrial Projects for Students (RIPS) program.

WORK EXPERIENCE

Graduate Student Instructor (ECE 567 Reinforcement Learning Theory) — Ann Arbor, MI
University of Michigan — Winter 2026

- Sole GSI for 100+ student graduate course; designed and delivered lectures; authored exams, problem sets, and rubrics.
- Modernized curriculum with new lectures and assignments on unsupervised RL and RL from human feedback.

PUBLICATIONS AND PREPRINTS

Motion-Planning via Contrastive Reinforcement Learning and Monte-Carlo Tree Search — RLC Workshop 2025
K. Kanarios and L. Ying.* — (link)
Cost Aware Best Arm Identification — RLC 2024
K. Kanarios, Q. Zhang, and L. Ying.* — (link)
Parallel Algebraic Multigrid for Fusion and Higher-Order PDEs — LLNL Technical Report 2023
S. Boileau, A. Das*, K. Kanarios*, and L. Krajcoviechova*.* — (link)

MENTORSHIP AND SERVICE

Reviewer. RLC (2025, 2026), L4DC (2025).
Founder and Organizer, Reinforcement Learning Seminar. Initiative to cultivate reinforcement learning community (*YouTube*).
Mentor, ECE Pure Program. Advise multiple projects on topics in reinforcement learning spanning both undergraduate and masters students.

Personal Interests: Running marathons, bowling (once bowled a 297), and dark roasts.